expression, can be adapted to the requirements of the general theory of relativity? The belief, prevailing among the physicists of today, that such an attempt would be hopeless, may have its root in the unjustifiable idea that such a theory should lead, as a first approximation, to the equations of classical mechanics for the motion of corpuscles, or at least to total differential equations. As a matter of fact up to now we have never succeeded in representing corpuscles theoretically by fields free of singularities, and we can, a priori, say nothing about the behavior of such entities. *One thing*, however, is certain: if a field theory results in a representation of corpuscles free of singularities, then the behavior of these corpuscles with time is determined solely by the differential equations of the field.

§ 6. Relativity Theory and Corpuscles

I shall now show that, according to the general theory of relativity, there exist singularity-free solutions of field equations which can be interpreted as representing corpuscles. I restrict myself here to neutral particles because, in another recent publication in collaboration with Dr. Rosen, I have treated this question in a detailed manner, and because the essentials of the problem can be completely shown by this case.

The gravitational field is entirely described by the tensor $g_{\lambda\nu}$. In the three-index symbols $\Gamma_{\lambda\nu}\sigma$ there appear also the contravariants $g^{\lambda\nu}$ which are defined as the minors of the $g_{\lambda\nu}$ divided by the determinant $g(= |g_{\alpha\beta}|)$. In order that the R_{ik}, shall be defined and finite, it is not sufficient that there shall be, for the environment of every part of the continuum, a system of coordinates in which the $g_{\lambda\nu}$ and their first differential quotients are continuous and differentiable, but it is also necessary that the determinant g shall nowhere vanish. This last restriction is, however, eliminated if one replaces the differential equations $R_{ik} = 0$ by $g^2R_{ik} = 0$, the left hand sides of which are *whole* rational functions of the g_{ik} and of their derivatives.

These equations have the centrally symmetrical solutions indicated by Schwarzschild

$$ds^2 = -\frac{1}{1-2m/r}\,dr^2 - r^2\,(d\theta^2 + \sin^2\theta\, d\varphi^2) + (1 - 2m/r)\,dt^2$$

This solution has a singularity at r = 2m, since the coefficient of dr^2 (i.e. g_{11}), becomes infinite on this hypersurface. If, however, we replace the variable *r* by ρ defined by the equation

$$\rho^2 = r - 2m$$

we obtain

$$ds^2 = -\frac{4\,(2m + \rho\rho^2)}{}\,d\rho^2 - (2m + \rho^2)^2\,(d\theta^2 + \sin^2\theta\, d\varphi^2) + \rho^2/2m + \rho^2\,dt^2$$

This solution behaves regularly for all values of p. The vanishing of the coefficient of dt^2 i.e. (g_{44}) for ρ = 0 results, it is true, in the consequence that the determinant g vanishes for this value; but, with the methods of writing the field equations actually adopted, this does not constitute a singularity. If ρ extends from - ∞ to +∞, then *r* runs from + ∞ to *r* = 2m and then back to +∞, while for such values of r as correspond to *r* < 2m there are no corresponding real values of p. Hence the Schwarzschild solution becomes a regular solution by representation of the physical space as consisting of two identical "shells" neighboring upon the hypersurface ρ = 0, that is *r* = 2m, while for this hypersurface the determinant *g* vanishes. Let us call such a connection between the two (identical) shells a "bridge." Hence the existence of such a bridge between the two shells in the finite realm corresponds to the existence of a material neutral particle which is described in a manner free from singularities.

The solution of the problem of the motion of neutral particles evidently amounts to the discovery of such solutions of the gravitational equations (written free of denominators), as contain several bridges.

The conception sketched above corresponds, a priori, to the atomistic structure of matter insofar as